

what are the roots of human obsession with justice?

The human preoccupation with justice originates from evolved psychological mechanisms for fairness and punishment, which emerge early in life and are modified by cultural practices.

Abstract

Human justice arises from interwoven biological, developmental, and cultural forces. Nine of ten studies argue that evolved predispositions—mechanisms for fairness, punishment, and cooperation—form the basis of justice sensitivity. Several sources report that moral sense emerges early in life, with evidence from developmental psychology and neuroscience showing infants and young children display fairness concerns, sometimes linked to empathy. Neurobiological investigations further reveal that cognitive, emotional, and motivational processes combine in distinct neural circuits that underlie justice-related evaluations. At the same time, cultural and institutional factors shape how these universal mechanisms are expressed and institutionalized, with at least six studies noting that local norms and historical contingencies modulate the basic human drive for justice.

Key points:

1. Evolved psychological traits promote fairness and punishment as adaptive strategies.
2. Early developmental evidence supports the emergence of justice intuitions.
3. Integrated neurobiological circuits underlie moral decision-making.
4. Cultural practices and institutions adapt universal justice mechanisms to local contexts.

Paper search

Using your research question "what are the roots of human obsession with justice?", we searched across over 126 million academic papers from the Semantic Scholar corpus. We retrieved the 50 papers most relevant to the query.

Screening

We screened in papers that met these criteria:

- **Human Psychological Focus:** Does the study examine human psychological, evolutionary, or neurobiological aspects of justice-seeking behavior (including neural correlates of justice-related decision-making)?
- **Sociocultural Analysis:** Does the research investigate cultural, sociological, or cross-cultural factors in justice development and perception?
- **Developmental Perspective:** Does the study examine either the emergence of justice concepts in children or aspects of moral development related to justice orientation?
- **Study Type:** Is the study either primary empirical research or a systematic review/meta-analysis of justice-related behavior and cognition?
- **Empirical Evidence:** Does the study present or analyze empirical evidence rather than purely theoretical arguments or opinions?
- **Human Focus:** Does the study focus on human behavior and cognition (rather than exclusively on animal behavior)?

- **Fundamental Focus:** Does the study examine fundamental aspects of justice orientation rather than focusing solely on specific procedures, policies, or legal systems?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Theoretical Framework:**

Identify the primary theoretical perspective or approach used in the study to explain the origins of justice. Look in the introduction, theoretical discussion, or conclusion sections.

Extract the specific theoretical lens, such as:

- Evolutionary psychology
- Neo-Darwinian theory
- Biological/genetic basis of social behavior
- Cultural institutional approach
- Psychological mechanisms perspective

If multiple perspectives are used, list them in order of prominence. If the perspective is not explicitly stated, note "Not clearly specified" and provide a brief description of the implied theoretical approach based on the text.

- **Key Conceptual Arguments:**

Identify and summarize the main conceptual arguments about the origins of justice.

Look for:

- Core hypotheses about how justice emerges
- Explanations of the relationship between biological and cultural factors
- Arguments about the psychological or social mechanisms underlying justice

Extract 2-3 key points that directly address the origins or development of justice. Use direct quotes if possible, or provide a concise paraphrase. If no clear arguments are found, note "No specific arguments identified."

- **Evidence Sources:**

Identify the types of evidence used to support arguments about the origins of justice:

- Evolutionary research
- Psychological studies
- Anthropological evidence
- Biological research
- Game theory experiments
- Historical analysis

List all types of evidence used, noting the primary source type. If no empirical evidence is presented, mark as "Theoretical/conceptual analysis only".

- **Comparative or Cross-Cultural Analysis:**

Determine if the study includes:

- Comparisons across different cultural contexts
- Analysis of cross-cultural variations in justice concepts
- Examination of universal vs. culturally specific aspects of justice

Extract specific details about comparative approaches, including:

- Cultures or societies compared
- Key differences or similarities identified
- Methodological approach to cross-cultural analysis

If no comparative analysis is present, note "No cross-cultural comparison conducted".

- **Novel Contributions:**

Identify the study's unique contributions to understanding the origins of justice:

- New theoretical insights
- Challenges to existing paradigms
- Innovative explanatory mechanisms

Look in the discussion, conclusion, or final sections for explicit statements about the study's novel contributions. Extract 1-2 key points that represent the most significant novel insights.

If no clear novel contributions are identified, note "No specific novel contributions highlighted".

Results

Characteristics of Included Studies

Study	Study Type	Research Focus	Methodology	Key Variables	Full text retrieved
Decety and Wheatley, 2015	Multidisciplinary review	Moral brain, origins of justice	Synthesis of empirical research from evolutionary biology, developmental psychology, neuroscience, anthropology	Evolution, development, neurobiological mechanisms, cultural influences	No

Study	Study Type	Research Focus	Methodology	Key Variables	Full text retrieved
Baumard and Chevallier, 2012	Empirical psychological study	Evolutionary roots of belief in immanent justice	Experimental studies on fairness and justice intuitions	Evolved sense of fairness, implicit justice beliefs, cultural success of justice beliefs	No
Robinson et al., 2006	Theoretical and empirical analysis	Shared intuitions of justice	Review of empirical evidence, cross-cultural and developmental studies	Evolved predisposition, universality, social learning vs. evolution	Yes
Decety and Yoder, 2017	Integrative review	Justice motivation	Synthesis of findings from evolution, psychology, behavioral economics, neuroscience	Heuristics, deliberation, neural circuitry, individual variation	No
Cushman, 2015	Theoretical review	Punishment, intuitions to institutions	Review of psychological, evolutionary, and institutional research	Adaptive history, institutional exaptation, psychological mechanisms	Yes
Eisner et al., 2017	Review	Criminal justice institutions, evolutionary psychology	Review of evolutionary, game theory, cross-cultural, and historical evidence	Cross-cultural variability, procedural fairness, legitimacy	Yes
Sznycer et al., 2021	Theoretical and empirical analysis	Justice-making institutions, adaptationist theory	Review and cross-cultural experiments	Conflict adaptations, universal principles, cultural parameters	Yes
Hatfield and Rapson, 2005	Social psychological review	Social justice, cultural conflict	Review of psychological and cultural factors in conflict	Moral disengagement, us-vs-them, cultural/religious influences	No

Study	Study Type	Research Focus	Methodology	Key Variables	Full text retrieved
O'Manique, 2002	Theoretical monograph	Evolution of morality, human rights, law	Neo-Darwinian theory (a modern evolutionary framework), evolutionary research, critique of Western paradigm	Empathy, mutual care, community, genetic inclinations	No
Chen et al., 2018	Review	Developmental origins of empathy, morality, justice	Review of developmental, neurobiological, and evolutionary studies	Empathy, early ontogeny, neurobiological precursors	Yes

Summary of Study Characteristics:

- Study Types:
 - 6 review articles (multidisciplinary, integrative, social psychological, theoretical)
 - 2 studies combining theoretical and empirical analysis
 - 1 empirical psychological study
 - 1 theoretical monograph
- Methodologies:
 - 8 studies used review or synthesis approaches
 - 2 studies included experimental methods (including cross-cultural experiments)
 - 4 studies included cross-cultural, developmental, or empirical evidence
 - 1 study used a theoretical/Neo-Darwinian/critique approach
- Key Variables Addressed:
 - Cultural influences, parameters, or variability: 6 studies
 - Evolution, evolved predisposition, or adaptive history: 5 studies
 - Fairness, justice beliefs, or procedural fairness: 4 studies
 - Empathy or mutual care: 3 studies
 - Neurobiological mechanisms, circuitry, or precursors: 3 studies
 - Universality or universal principles: 2 studies
 - Other variables (each in 1 study): social learning vs. evolution, moral disengagement/us-vs-them, legitimacy, heuristics/deliberation, institutional exaptation, conflict adaptations, community, individual variation, early ontogeny
- Full Text Availability:

- 5 studies had full text available
- For studies without full text, findings are based on abstracts
- Notable Pattern:
 - None of the included studies used only quantitative experimental methods without a review or theoretical component

Thematic Analysis

Evolutionary Foundations

Study	Evolutionary Arguments	Evidence Base	Universality/Adaptation
Decety and Wheatley, 2015	Moral cognition evolved to aid cooperation; justice as an adaptive trait	Evolutionary biology, cross-species comparisons	Universal mechanisms shaped by evolution
Baumard and Chevallier, 2012	Belief in immanent justice rooted in evolved fairness	Psychological experiments, evolutionary theory	Implicit, widespread across individuals
Robinson et al., 2006	Shared intuitions of justice as evolved predispositions	Animal behavior, developmental psychology, cross-cultural studies	Universality across cultures and development
Decety and Yoder, 2017	Justice motivation as a product of evolution and neural heuristics	Evolutionary, psychological, neuroscientific evidence	Early, universal sensitivity to justice
Cushman, 2015	Punishment as an evolved mechanism for social regulation	Evolutionary, psychological, anthropological evidence	Universal psychological mechanisms
Eisner et al., 2017	Criminal justice rooted in evolved psychology, but variable	Evolutionary, game theory, cross-cultural evidence	Some universals, but significant variability
Sznycer et al., 2021	Justice institutions built on ancient conflict adaptations	Evolutionary, psychological, cross-cultural experiments	Universal principles, local parameters
O'Manique, 2002	Justice from genetically-based inclinations, empathy	Neo-Darwinian theory (a modern evolutionary framework), evolutionary research	Emphasis on positive, community-based adaptation
Chen et al., 2018	Justice emerges from early altruism, cooperation	Evolutionary, developmental, neurobiological studies	Early, universal emergence

Study	Evolutionary Arguments	Evidence Base	Universality/Adaptation
Hatfield and Rapson, 2005	Not evolutionary-focused	Social psychology, cultural analysis	Focus on psychological/cultural mechanisms

Key Patterns Across Studies:

- Evolutionary Basis:
 - 9 of 10 studies proposed an evolutionary basis for justice, with diverse arguments including evolved cooperation, fairness, predispositions, motivation, punishment, conflict adaptation, genetic inclinations, and early altruism.
 - 1 study (Hatfield and Rapson, 2005) did not focus on evolutionary explanations, instead emphasizing social and cultural mechanisms.
- Evidence Base:
 - Evolutionary biology/theory/research: 7 studies
 - Psychological experiments/evidence: 4 studies
 - Cross-species or animal behavior: 2 studies
 - Developmental psychology/studies: 2 studies
 - Cross-cultural studies/experiments: 3 studies
 - Neuroscientific or neurobiological evidence: 2 studies
 - Anthropological evidence, game theory, Neo-Darwinian theory, social psychology, and cultural analysis: each in 1 study
- Universality or Adaptation Claims:
 - 6 studies made explicit claims of universality or widespread presence of justice-related mechanisms
 - 2 studies described partial universality or universality with local variation
 - 1 study emphasized adaptation or community-based mechanisms
 - 1 study focused on psychological/cultural mechanisms without evolutionary universality claims

Developmental Trajectory

Study	Developmental Arguments	Evidence Base	Role of Empathy
Decety and Wheatley, 2015	Early developmental emergence of moral sensibility	Developmental psychology, neuroscience	Empathy as a component of moral cognition
Robinson et al., 2006	Predictable stages in moral judgment from infancy	Child development literature	Sociality and empathy as drivers

Study	Developmental Arguments	Evidence Base	Role of Empathy
Decety and Yoder, 2017	Justice sensitivity present from a young age	Developmental, psychological, neuroscientific evidence	Empathy and neural heuristics
Chen et al., 2018	Sensitivity to fairness in infants as young as 15 months	Developmental, neurobiological studies	Empathy, harm sensitivity, moral appraisal
O'Manique, 2002	Justice constructed within communities post self-consciousness	Theoretical, evolutionary	Empathy and mutual care central
Baumard and Chevallier, 2012	Implicit justice intuitions operate early	Psychological experiments	Evolved fairness, not explicit empathy
Hatfield and Rapson, 2005	Psychological mechanisms develop in cultural context	Social psychology	Empathy, but also moral disengagement
Cushman, 2015	Punishment judgments develop from evolved mechanisms	Psychological, developmental evidence	No mention found
Eisner et al., 2017	No mention found	Review	No mention found
Sznycer et al., 2021	No mention found	Review, experiments	No mention found

Key Patterns Across Studies:

- Developmental Arguments:
 - 5 studies described early or emergent moral sensibility, justice, or fairness in development
 - 1 study described a stage-based progression in moral judgment
 - 2 studies argued for the construction of justice or moral mechanisms within communities or cultural contexts, often after self-consciousness develops
 - 2 studies did not mention developmental arguments
- Role of Empathy:
 - Empathy was central or explicit in 2 studies
 - 4 studies described empathy as one of several factors in moral development
 - 1 study focused on evolved fairness or justice mechanisms without explicit reference to empathy
 - 3 studies did not mention empathy as a primary focus
- Evidence Base:
 - Developmental psychology, neuroscience, or child development literature: 5 studies
 - Theoretical or evolutionary approach: 1 study
 - Psychological experiments: 1 study
 - Social psychology: 1 study
 - Reviews or experiments: 2 studies

Neurobiological Mechanisms

Study	Neurobiological Focus	Evidence Base	Integration of Processes
Decety and Wheatley, 2015	Brain circuits for moral decision-making	Neuroscience, social neuroscience	Cognitive, emotional, motivational integration
Decety and Yoder, 2017	Distinct/interacting neural circuitry for justice motivation	Neuroscience, psychology	Rapid heuristics and deliberation
Chen et al., 2018	Neurobiological precursors of empathy, morality, justice	Neuroimaging, developmental neuroscience	Affective arousal, perspective-taking, emotional sharing
Robinson et al., 2006	Brain science supports evolutionary basis	Neuroscience, developmental studies	Biological underpinnings of moral judgment
Baumard and Chevallier, 2012	No mention found	Psychological experiments	No mention found
Cushman, 2015	Psychological mechanisms, some neural basis	Review	No mention found
Eisner et al., 2017	Neuroscientific studies referenced	Review	No mention found
Sznycer et al., 2021	Emotion systems as adaptations	Neuroscience, psychology	Integration with cognitive mechanisms
O'Manique, 2002	Biological/genetic inclinations	Evolutionary, theoretical	No detailed mention found
Hatfield and Rapson, 2005	Emotional processes in conflict	Social psychology	Cognitive-emotional interplay

Key Patterns Across Studies:

- Neurobiological Focus:
 - 8 studies had a primary neurobiological focus, including brain circuits, neural circuitry, neurobiological precursors, evolutionary/biological/genetic bases, emotion systems, and emotional processes
 - 2 studies did not have a primary neurobiological focus or only referenced neuroscientific studies
- Integration of Processes:
 - 7 studies addressed integration of cognitive, emotional, or motivational processes
 - * Cognitive integration or mechanisms: 4 studies
 - * Emotional or affective integration: 4 studies
 - * Motivational integration: 3 studies
 - * Other forms: perspective-taking, emotional sharing, heuristics/deliberation, biological underpinnings, cognitive-emotional interplay (each in 1 study)
 - 3 studies did not focus on integration of processes, or the integration was not detailed
- Evidence Base:

- Neuroscience: 4 studies
- Neuroimaging or developmental neuroscience: 2 studies
- Psychology or social psychology: 4 studies
- Evolutionary or theoretical approaches: 2 studies
- Review articles: 2 studies
- Psychological experiments: 1 study

Cultural-Institutional Evolution

Study	Institutional/Cultural Focus	Evidence Base	Universal vs. Culture-Specific
Cushman, 2015	Institutions exapt evolved punishment mechanisms	Theoretical, anthropological, historical	Shared mechanisms, cultural variation in institutions
Eisner et al., 2017	Criminal justice institutions reflect evolved psychology, but vary	Cross-cultural, historical, game theory	Universal features, significant cross-cultural variability
Sznycer et al., 2021	Justice institutions built on adaptations, with local parameters	Cross-cultural experiments, adaptationist theory	Universal principles, local cultural variation
Robinson et al., 2006	Shared intuitions, but social conventions vary	Cross-cultural, developmental studies	Universal core, cultural differences in application
Decety and Wheatley, 2015	Culture shapes moral sensibility	Multidisciplinary review	Both universal and culture-specific elements
Baumard and Chevallier, 2012	Cultural success of justice beliefs	Psychological experiments	Universal intuitions, cultural expression
O'Manique, 2002	Justice constructed in communities, transcending biology	Theoretical, evolutionary	Emphasis on community/culture
Hatfield and Rapson, 2005	Cultural, historical, religious factors in justice/injustice	Social psychology, historical analysis	Focus on cultural specificity
Decety and Yoder, 2017	No mention found	Integrative review	No mention found
Chen et al., 2018	No mention found	Developmental, neurobiological	No mention found

Key Patterns Across Studies:

- Universal vs. Culture-Specific Findings:
 - 6 studies identified universal mechanisms or principles underlying justice or punishment, but also reported substantial cultural variation in their expression or institutionalization
 - 1 study emphasized community or culture as the primary basis for justice, without highlighting universality

- 1 study focused on cultural specificity only
 - 2 studies did not mention universality or cultural specificity as a primary focus
 - Institutional/Cultural Focus:
 - 3 studies primarily focused on institutions (such as criminal justice or evolved punishment mechanisms)
 - 4 studies primarily focused on culture or community
 - 1 study addressed both institutions and culture
 - 2 studies did not have a primary focus on institutions or culture
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Summary

The included studies report that the roots of human obsession with justice are addressed through several intersecting themes:

- **Evolutionary Foundations:** Most studies propose that justice-related intuitions and motivations have evolutionary origins, with some emphasizing universality and others highlighting adaptation to local contexts.
- **Developmental Trajectory:** Multiple studies describe the early emergence of moral sensibility and justice in development, with empathy playing a variable but often important role.
- **Neurobiological Mechanisms:** Many studies focus on the neural and biological substrates of justice, often emphasizing the integration of cognitive, emotional, and motivational processes.
- **Cultural-Institutional Evolution:** While some universal mechanisms are reported, the expression of justice in institutions and cultural practices is highly variable, shaped by local norms, historical contingencies, and social structures.

Overall, the evidence from these studies suggests that the human obsession with justice is shaped by a complex interplay of evolved psychological mechanisms, developmental processes, neurobiological substrates, and cultural-institutional adaptations. The findings are based on a mix of review articles, theoretical analyses, and empirical studies, with most evidence coming from reviews and syntheses rather than large-scale quantitative experiments.

References

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